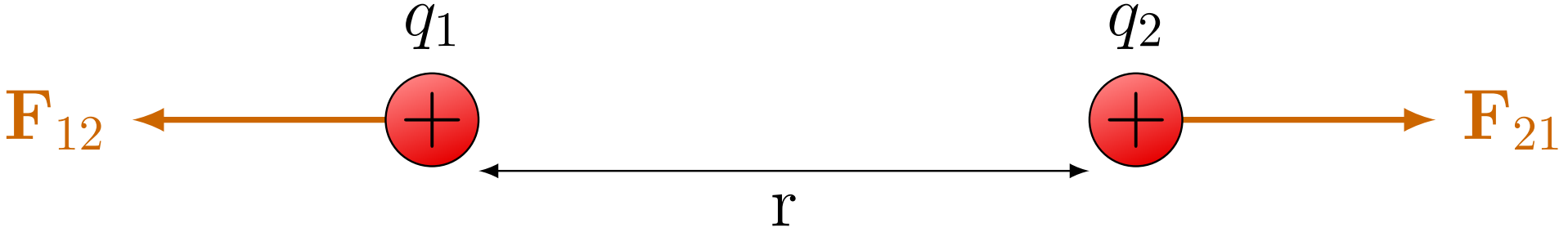
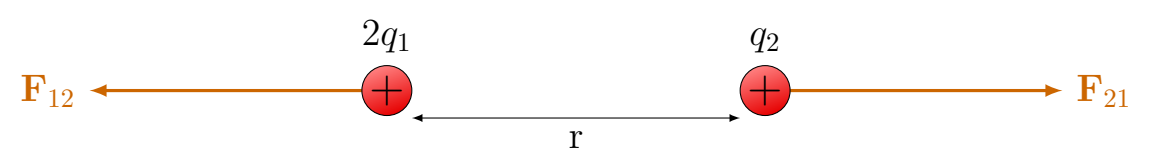
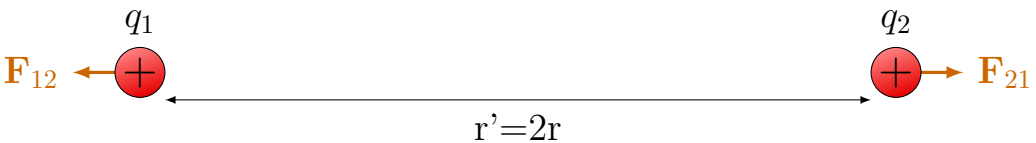
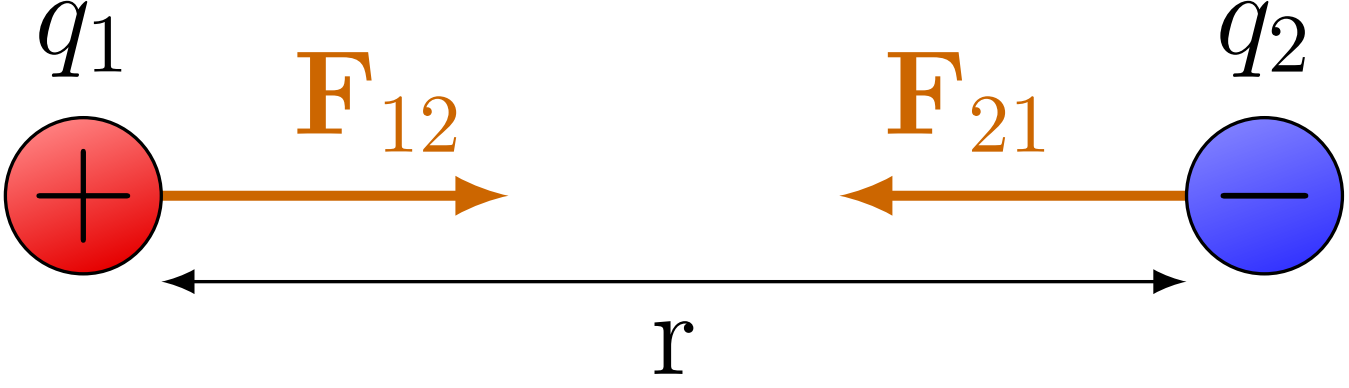
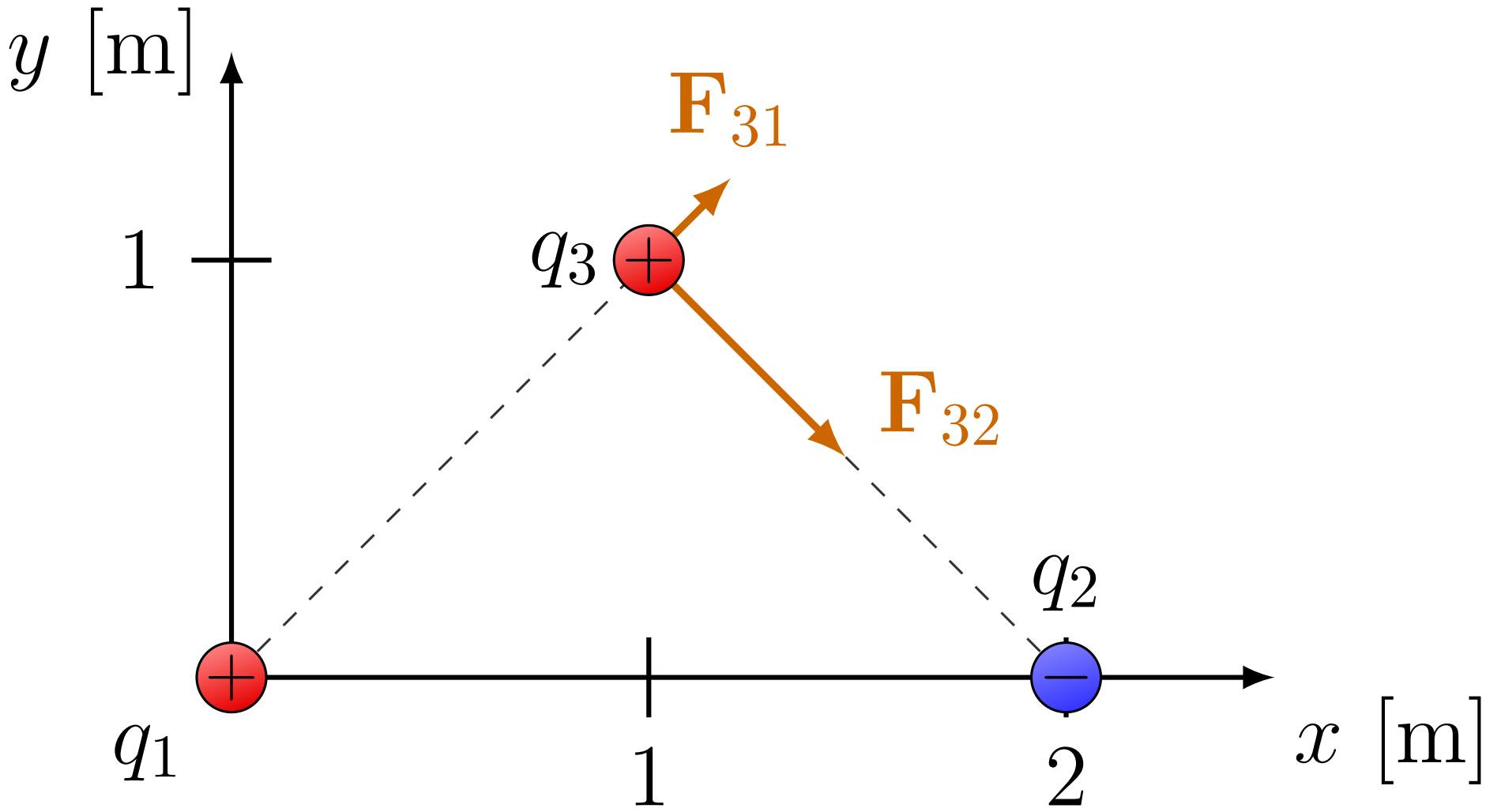

q_1
q_2
$\mathbf{F}_{12}$
$\mathbf{F}_{21}$
r


$2q_1$
q_2
$\mathbf{F}_{12}$
$\mathbf{F}_{21}$
r


q_1
q_2
$\mathbf{F}_{12}$
$\mathbf{F}_{21}$
r'=2r


q_1
$\mathbf{F}_{12}$
$\mathbf{F}_{21}$
q_2
r


y [m]
1
q3
F31
F32
q2
q1
1
2
x [m]